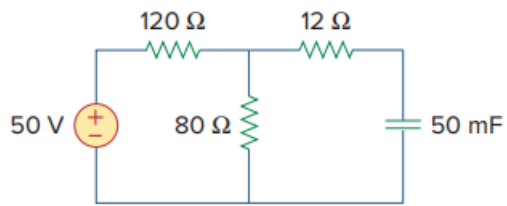


Problem

Find the time constant for the RC circuit in Fig. 7.82.

Fig. 7.82:



Step-by-step solution

Step 1 of 6

Refer to Figure 7.82 in the textbook.

Consider the following circuit diagram as shown in Figure 1.

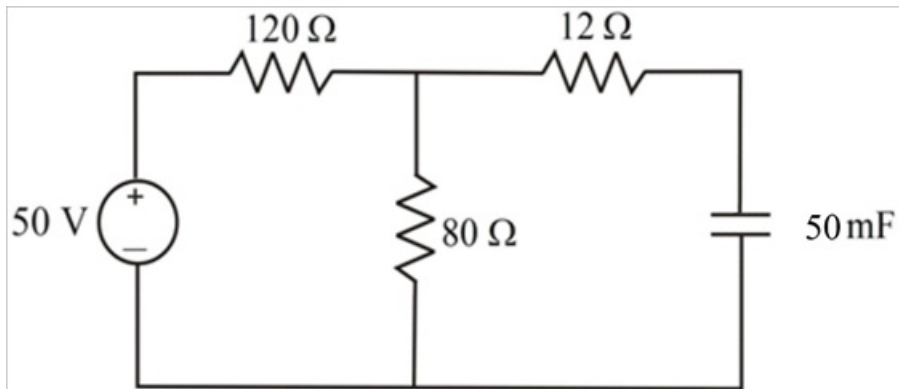
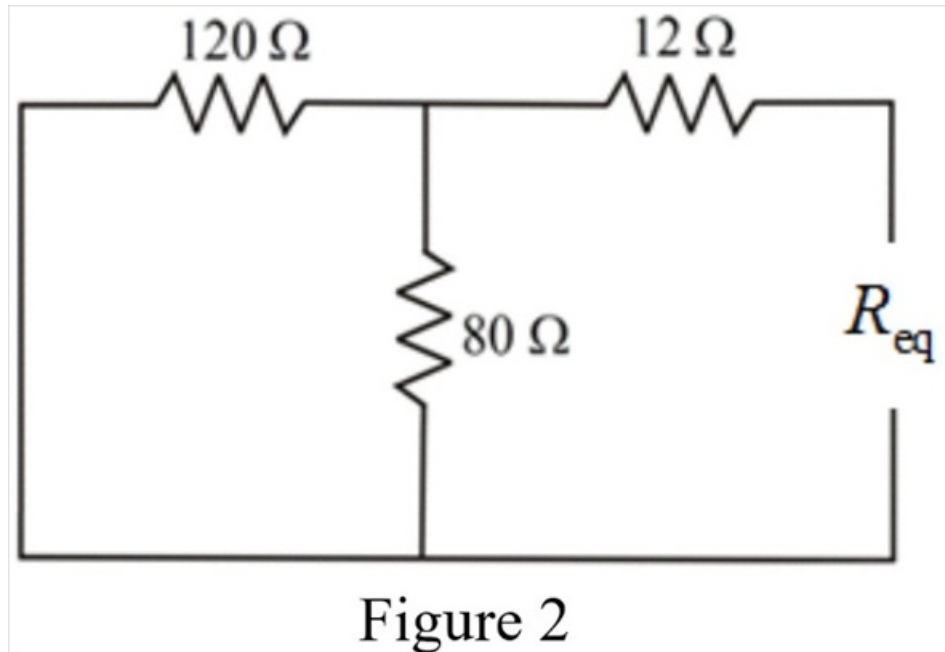


Figure 1

Step 2 of 6

To calculate the equivalent resistance, short the voltage source to zero resistance and open the capacitor. Draw the modified circuit diagram as shown in Figure 2.



Step 3 of 6

Resistors 120 Ω and 80 Ω in the Figure 2, are connected in parallel.

Determine the equivalent resistance of the parallel resistors.

$$\begin{aligned} R &= (120) \parallel (80) \\ &= \frac{(120)(80)}{120 + 80} \\ &= \frac{9600}{200} \\ &= 48 \Omega \end{aligned}$$

Step 4 of 6

Draw the modified circuit after converting parallel resistance into equivalent resistance as shown in Figure 3.

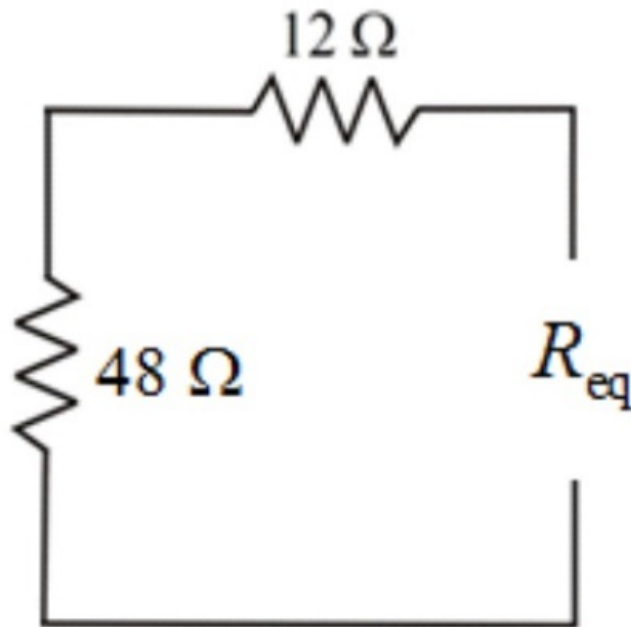


Figure 3

Step 5 of 6

Looking from the open terminals, resistors $48\ \Omega$ and $12\ \Omega$ in the Figure 3, are connected in series.

Determine the equivalent resistance.

$$\begin{aligned} R_{eq} &= 48 + 12 \\ &= 60\ \Omega \end{aligned}$$

Step 6 of 6

Write an expression for the time constant τ .

$$\tau = R_{eq} C$$

Substitute $60\ \Omega$ for R_{eq} and $50\ \text{mF}$ for C in the time constant expression.

$$\begin{aligned} \tau &= (60)(50 \times 10^{-3}) \\ &= 3\ \text{s} \end{aligned}$$

Therefore, the time constant, τ of the RC circuit is 3 s.